

Object-centric Sampling for 3D Object Detection via Pseudo Labels

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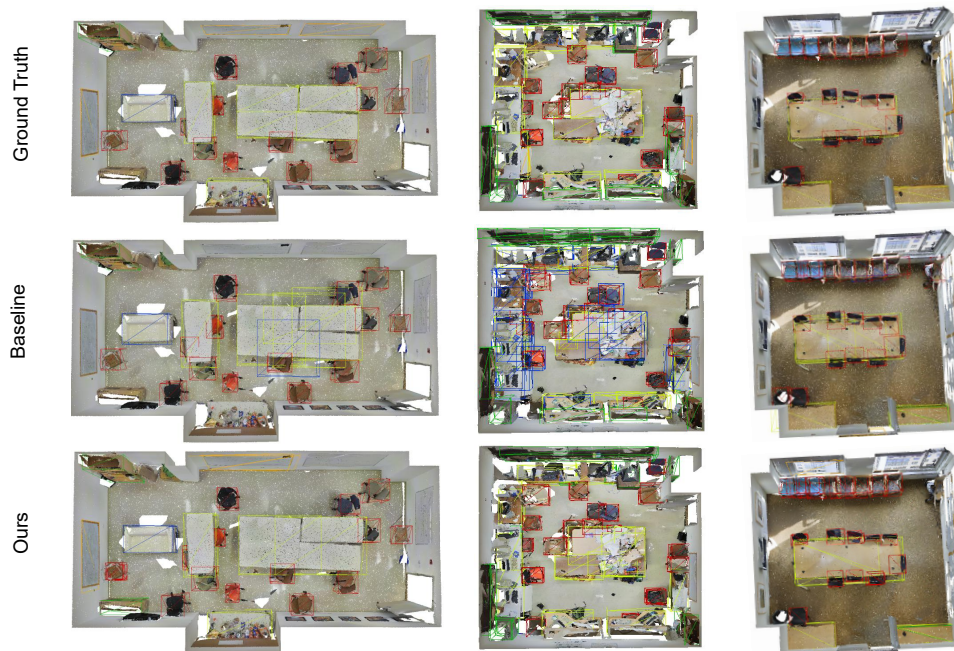


Figure 1: Qualitative results on the S3DIS dataset. Compared to the baseline, our Object-Centric Sampling Module generates predictions that closely match the ground truth.

ABSTRACT

Point cloud-based 3D object detection has become a core technology for 3D scene understanding in various fields, including autonomous driving, robotics, and augmented reality. In indoor 3D object detection, random sampling is commonly applied to input points to control the model's input size. However, this approach often fails to preserve sufficient object points, which typically occupy a much smaller proportion compared to background points such as walls and floors. This randomness leads to performance instability and variability across different trials. To address these issues, we propose a pseudo label-based Object-Centric Sampling Module. Specifically, we utilize a pretrained segmentation model to obtain panoptic segmentation-based pseudo labels and determine the sampling ratio between object and non-object points through a weighted average strategy. This design maximizes the preservation of object information while maintaining an appropriate amount of background points necessary for capturing the global context of the scene. Overall, our method effectively addresses the limitations of random sampling in point cloud-based 3D object detection.

Index Terms: 3D object detection, indoor scene understanding, point clouds

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1 INTRODUCTION

3D object detection is one of the key computer vision tasks for 3D scene understanding, which involves predicting object bounding boxes from point clouds. Research in indoor 3D object detection has proposed methods that leverage anchors and single-pass processing to achieve real-time inference, as well as approaches that can be applied across multiple domains [1, 2]. In addition, some studies have explored the use of superpoints to simultaneously improve computational efficiency and detection performance [3].

ScanNet [4] and S3DIS [5] are representative benchmarks for indoor 3D object detection, providing high-density point clouds with hundreds of thousands of points per scene. The pioneering VoteNet [6] applied random sampling to augment the training data. 40K points per scene from ScanNet [4] were used for training and inference in VoteNet [6]. FCAF3D [7] indicated that using 100K points for random sampling led to higher performance compared to 20K and 40K points. Therefore, the number of sampled points was fixed at 100K. In UniDet3D [2], random sampling is not applied for ScanNet [4], whereas S3DIS [5] uses 180K points per scene. Despite the development of various indoor 3D object detection approaches, point sampling remains necessary for datasets such as S3DIS [5], where each scene contains a very large number of points. For comparing the performance of 3D object detection models trained on such datasets, the mAP metric is evaluated in two ways: the average over 25 trials and the best value. This suggests that the gap between the worst-case and average can be as large as the gap between the best and average. We attribute this variability to the stochasticity introduced by random sampling and aim to address this issue.

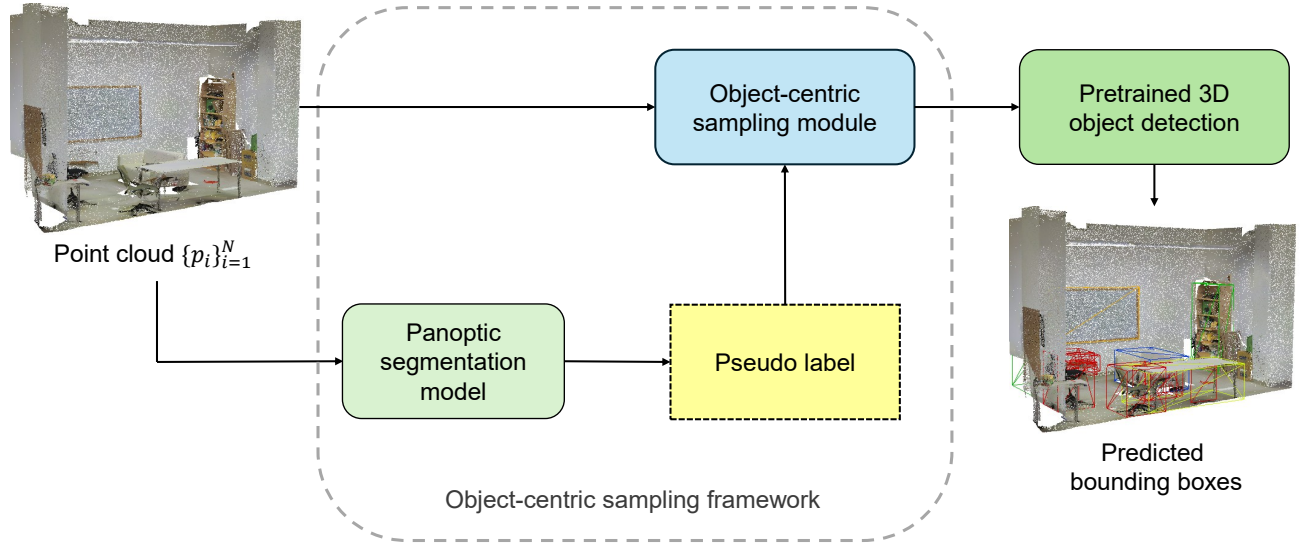


Figure 2: Overview of our framework. Given a point cloud, we obtain pseudo labels through panoptic segmentation. The pseudo labels and the input point cloud are then fed together into the Object-Centric Sampling Module, which outputs an object-centric sampled point cloud. Finally, the sampled point cloud is used as the input to the 3D object detection model.

Random sampling treats all points in a scene with equal importance. In S3DIS [5], points belonging to objects are relatively few, while most points correspond to background elements such as walls and floors. Point sampling has been widely used and can help reduce computational cost. However, objects with relatively few points retain even fewer points after random sampling. As a result, we observe that insufficient feature extraction for objects leads to suboptimal detection performance. Motivated by this observation and the importance of preserving object information in object detection [8], we propose an object-centric sampling strategy.

In this paper, we propose an Object-Centric Sampling Module based on pseudo labels to address the aforementioned problems. Specifically, we obtain the ratio of object and non-object points through pseudo labels and determine the sampling ratio of object points by applying a weighted average strategy. Furthermore, to handle cases where the number of object points is insufficient, we introduce an exponential decay-based weighting strategy. This module replaces conventional random sampling in 3D object detection with object-centric sampling. Experimental results on the S3DIS [5] dataset demonstrate that applying our module to UniDet3D [2] improves performance in terms of mAP@0.25 and mAP@0.5. Fig. 1 presents qualitative results corresponding to the quantitative evaluation. In addition, the proposed method significantly reduces performance variability across repeated experiments, thereby enhancing model reliability. This study overcomes the limitations of random sampling in indoor 3D object detection and demonstrates that preserving object-centric information contributes to improved model robustness.

2 RELATED WORK

2.1 3D Point Cloud Indoor Object Detection

3D object detection has evolved along three major paradigms: voting-based, dense 3D CNN-based, and transformer-based methods. Following the pioneering VoteNet [6], which established the Hough Voting mechanism, various voting-based models [9, 10, 11, 12, 13, 14, 15] have been proposed. In addition, SPGroup3D [3] adopts an approach that combines superpoints with voting. Meanwhile, dense 3D CNN-based strategies [7, 16, 17] utilize anchors

or voxels. These methods achieve real-time inference performance. SR3D [1] further mitigates the gap between training and inference. Transformer-based models [18, 19, 20] have advanced by incorporating attention mechanisms to effectively capture global contextual information. Recently, research has expanded to domains such as UniDet3D [2], UniDETR [21].

However, despite these architectural advances, most existing methods still rely on random sampling strategies during training and inference. In particular, random sampling during inference can lead to inefficient point utilization and unstable detection performance. This observation highlights the importance of sampling strategies in 3D object detection.

2.2 3D Point Cloud Indoor Panoptic Segmentation

Panoptic segmentation has been extensively studied using RGB images in prior works [22, 23, 24]. These approaches leverage rich color and texture information to achieve accurate scene understanding. However, OneFormer3D [25] performs panoptic segmentation using only point cloud data without relying on RGB images. UniSeg3D [26] incorporates additional modalities such as click and text inputs to enhance segmentation performance. Furthermore, UNIT3D [27] integrates detection-based geometric information into the segmentation framework, leading to more precise estimation of Panoptic Quality for Things. We aim to enhance sampling efficiency by exploiting the capabilities of indoor panoptic segmentation.

3 METHOD

3.1 Overview

Fig. 2 presents an overview of our framework. Our framework introduces an object-centric sampling strategy that utilizes semantic priors to effectively sample point clouds while retaining object details. We use UniDet3D [2] as the object detection baseline model and OneFormer3D [25] as an auxiliary segmentation model. We use a point cloud consisting of N points as input, which is defined as $\{p_i\}_{i=1}^N$. Each point p_i is composed of spatial coordinates (x_i, y_i, z_i) and color (r_i, g_i, b_i) . This point cloud is fed into OneFormer3D [25] to obtain pseudo labels. These labels consist of instance and

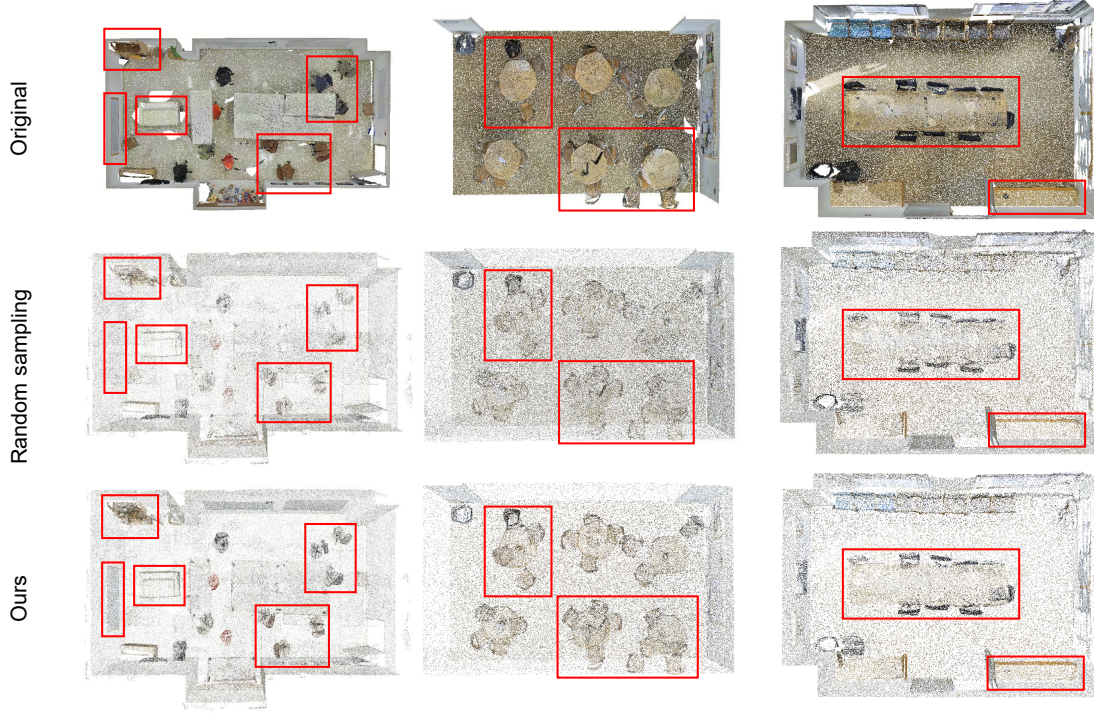


Figure 3: We visualize the results of random sampling and the output of our object-centric sampling module on S3DIS with 180K sampled points. From the visualization results, as shown in the red boxes, it can be observed that our sampling method captures more object points compared to random sampling. Even in regions outside the boxes, the sampled points are more concentrated on object areas rather than non-object regions.

semantic information, represented as $\{\{ins_i\}_{i=1}^N, \{sem_i\}_{i=1}^N\}$. In the Object-Centric Sampling Module, pseudo labels are used to distinguish object points, after which the object-centric approach is applied to generate a sampled point cloud. Finally, the sampled point cloud is used as input to the object detection baseline model.

3.2 Pseudo Labeling

Given an input point cloud, it is first input to OneFormer3D [25], a panoptic segmentation model, to generate pseudo labels. The pseudo labels are obtained in the form of instance and semantic information, denoted as $\{\{ins_i\}_{i=1}^N, \{sem_i\}_{i=1}^N\}$. By obtaining these pseudo labels, we establish fundamental information for distinguishing between objects and non-objects.

3.3 Object-Centric Sampling Module

To replace the inherent random sampling in the object detection model with object-centric sampling, the aforementioned pseudo labels are employed. Using the semantic information, points are classified into object and non-object categories. Based on object points P_{obj} and non-object points $P_{non-obj}$, the sampling ratio for the down-sampled points is calculated as follows:

$$r = r_{obj} \cdot w + r_{non-obj} \cdot (1 - w), \quad (1)$$

$$w = e^{-2x}, \quad (2)$$

$$r_{max} = \max(r, r_{obj}), \quad (3)$$

where r_{obj} and $r_{non-obj}$ denote the proportion of object and non-object points in the entire pseudo label set. We perform object-centric sampling, but we ensure that non-object points are not completely ignored. Therefore, as defined in Equation (1), the ratio is

calculated as a weighted average of r_{obj} , following the standard statistical definition of a weighted mean. In Equation (2), the weight w is defined as an e^{-2x} function. This choice is motivated by the function e^{-kx} . The function is designed such that smaller values of x receive higher weights, and the assigned weights decrease progressively as x increases. To utilize the value r_{obj} , r_{obj} is used as the input x of the function. Since $r_{non-obj} = 1 - r_{obj}$ in Equation (1), the function becomes symmetric with respect to $r_{obj} = 0.5$. When $r_{obj} < 0.5$, the resulting ratio satisfies $r > r_{obj}$, which is not problematic. However, when $r_{obj} \geq 0.5$, the resulting ratio becomes $r \leq r_{obj}$. Therefore, to maintain the object-centric property, Equation (3) is used to prevent the ratio from decreasing. The final value of the object sampling ratio α to be used is as follows:

$$\alpha = \min(r_{max}, R), \text{ where } R = \frac{|P_{obj}|}{N_s}. \quad (4)$$

Here, $|P_{obj}|$ denotes the number of object points obtained from the pseudo labels, and $\text{num_sampling}(N_s)$ is the number of points to be sampled. Equation (4) is preferable for computing the sampling ratio; however, there may be cases where the number of pseudo object points is too small to satisfy the desired ratio. Given the resulting object sampling ratio α and non-object sampling ratio $1 - \alpha$, the ratio of points to be sampled for each object class r_c can be calculated as follows:

$$r_c = \frac{\alpha \cdot N_s}{|P_{obj}|}. \quad (5)$$

Using Equation (5), r_c is applied to perform random sampling for each object instance class. The same procedure is applied to non-object points, so that a total of N_s object-centric sampling points can be obtained.

Method	mAP ₂₅			mAP ₅₀			Num of sampling	Sampling time(s)
	Best	Avg	Diff	Best	Avg	Diff		
UniDet3D(w/ random sampling)	76.86	76.16	0.7	67.81	65.94	1.87	180K	6.13
UniDet3D(w/ ours)	77.87	78.45	0.58	68.6	69.39	0.79		31.55
SPGroup3D(w/ random sampling)	69.84	68.88	0.96	47.73	47.18	0.55	100K	5.59
SPGroup3D(w/ ours)	69.57	69.08	0.49	46.13	45.19	0.94		30.73
TR3D(w/ random sampling)	71.45	70.73	0.72	48.51	44.51	4.00	100K	5.05
TR3D(w/ ours)	74.85	73.94	0.91	49.85	48.94	0.91		30.74

Table 1: 3D object detection results on the S3DIS validation set. The main comparison is based on the best results from multiple experiments and the average over five trials. The difference between the best and average results is reported as “Diff”. “Num of sampling” denotes the number of points used for sampling, and “sampling time” indicates the time required to perform sampling using the random strategy and our method.

3.4 Integration with 3D Object Detection

The object-centric sampled point cloud obtained from the Object-Centric Sampling Module is used as input to the baseline model, UniDet3D [2]. In this process, the sampled points are directly fed into the model encoder without undergoing the random sampling procedure inherent in the object detection model. As a result, the model ultimately produces predicted bounding boxes for the objects.

4 EXPERIMENTS

4.1 Dataset and Evaluation

We evaluate our approach on the indoor benchmark S3DIS [5]. S3DIS [5] contains 272 rooms annotated with instance and semantic labels for five object categories. For validation, we use 68 rooms based on the official Area 5 split. We adopt mean average precision (mAP) under IoU thresholds of 0.25 and 0.5 as the evaluation metric. We report both the best and average performance across five trials. For the baseline and other object detectors, each model is trained five times and evaluated five times per training run, resulting in a total of 25 evaluations, and the average value is reported. However, since we use a pretrained model, we conduct only five evaluations and report the corresponding results.

For object-centric sampling, we adopted the same number of sampling points as the random sampling used in the original models. UniDet3D [2] sampled 180K points, whereas SPGroup3D [3] and TR3D [17] sampled 100K points each. In our experimental setup, we implement UniDet3D [2] and OneFormer3D [25] in the mmdetection3d [28] framework. All experiments are conducted using a single NVIDIA GeForce RTX 2080 Ti.

4.2 Main Results

We conducted experiments using SPGroup3D [3], TR3D [17], and UniDet3D [2] on the S3DIS [5] dataset. To ensure reliability, each model was evaluated over five trials. We report the Best and Average (Avg) performance, along with the Difference (Diff) between them, to quantify stability. Tab. 1 shows 3D object detection results on the validation set of the S3DIS [5] dataset. Regarding performance superiority, our method outperforms the baselines in most metrics. Specifically, for UniDet3D [2] and TR3D [17], we achieved higher scores in both Best and Average metrics. In the case of SPGroup3D [3], we observed an improvement in Average mAP@0.25, demonstrating that our method effectively raises the overall performance baseline.

In terms of stability, our method demonstrates more consistent performance, as evidenced by the reduced gap between the best and average scores. We successfully minimized the Difference for UniDet3D [2] across all metrics and significantly improved stability for SPGroup3D [3] on mAP@0.25. Notably, for TR3D [17]

on mAP@0.50, we drastically reduced the ‘Diff’ value (from 4.00 to 0.91), which demonstrates the robustness of our approach. Although a marginal increase in variability was observed for TR3D [17] on mAP@0.25, the substantial gains in overall performance validate the effectiveness of our method.

Fig. 1 and Fig. 3 compare the results of random sampling and object-centric sampling using our Object-Centric Sampling Module. Fig. 3 presents a visual comparison between random sampling and the proposed Object-Centric Sampling Module. The regions marked by red boxes indicate the target objects to be detected. From a visual perspective, it can be observed that our sampling method focuses more on objects and samples a larger number of object points. In addition, from the perspective of non-object points, including walls and floors, the proportion of non-object points decreases, while the proportion of object points increases. As a result, objects appear more clearly compared to those obtained through random sampling.

5 CONCLUSION

We address the limitations of random sampling in indoor 3D object detection, where the scarcity of object points can lead to information loss and unstable performance. To mitigate this, we propose an Object-Centric Sampling Module that leverages pseudo labels from the pretrained panoptic segmentation model to preserve object points. By dynamically adjusting the sampling ratio, our module consistently captures critical object features while retaining sufficient background context for global scene understanding. Experiments on S3DIS show that integrating our module into UniDet3D improves both mAP@0.25 and mAP@0.5. It also reduces performance variance, demonstrating the effectiveness of object-centric sampling over random strategies. These results highlight the potential of leveraging semantic priors for more robust and reliable 3D perception.

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